

Software Engineering

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What Will You Learn Today?

- SQA Reviews
 - Objectives of Reviews
 - Kinds of Reviews
- Responsibilities of Roles
 - Facilitator
 - Author
 - Recorder
 - Reviewer
 - Observer

Software ***Inspections*** are a disciplined engineering practice for detecting and correcting defects in software artifacts, and preventing their leakage into field operations.

In a ***review*** , a work product is examined for defects by individuals other than the person who produced it. A ***Work Product*** is any important deliverable created during the requirements, design, coding, or testing phase of software development. Research shows that reviews are one of the best ways to ensure quality requirements, giving you as high as a 10 to 1 return on investment. Reviews help you to discover defects and to ensure product compliance to specifications, standards or regulations

In a ***Walkthrough***, the producer describes the product and asks for comments from the participants. These gatherings generally serve to inform participants about the product rather than correct it.



SQA Reviews

What is a Review?

- A **process** or meeting during which a **work product**, or a set of work products, is presented to project **personnel**, **managers**, **users**, or other **interested parties** for **comment** or **approval**. Types include **code** review, **design** review, **formal qualification** review, **requirements** review, **test readiness** review

[IEEE Std. 610.12-1990]

Objectives of Reviews

- Identify required **improvements** in a product
- Assure that the **deliverable** is **complete**
- Assure that the **deliverable** is **technically correct**
- Measure the **progress** of the project

Objectives of Reviews

- Identify any defects early, thus resulting in cost and time savings
- Assure the quality of deliverable before the development process is allowed to continue
- Once a deliverable has been reviewed, revised as necessary, and approved, it can be safely used as a basis for further development

Kinds of Reviews

- Business reviews
- Technical reviews
- Management reviews
- Walk-throughs
- Inspections

Objectives of Business Reviews

- The deliverable is complete
- The deliverable provides the information required for the next phase
- The deliverable is correct

Objectives of Technical Reviews

- Point out needed **improvements** in the product of a single person or a team
- Confirm those **parts** of a **product** in which **improvement** is either **not desired** or **not needed**
- Software reviews are a **“filter”** for software engineering process

Objectives of Technical Reviews

- Reviews are **applied** at several points during software development and serve to **uncover errors** and **defects** that can then be removed
- Software reviews “**purify**” the software engineering activities

Objectives of Technical Reviews

- **Another reason** we need technical reviews is that although people are good at catching some of their own errors, **large classes of errors escape** the **originator** more easily than they escape anyone else

Objectives of Management Reviews

- Validate from a management perspective that the **project** is **making progress** according to the **project plan**
- Ensure a **deliverable** is **ready** for management **approval**
- **Resolve issues** that require management's attention



Review Roles

Review Roles

- Facilitator
- Author
- Recorder
- Reviewer
- Observer

Responsibilities of Facilitator

- Responsible for providing the **background** of the **work** and **assigning roles** to attendees
- Encourages all **attendees** to **participate**
- Keeps the **meeting focused** and **moving**
- Responsible for **gaining consensus** on **problems**



Responsibilities of Author

- Responsible for the **readiness** and **distribution** of **material** to be reviewed
- During the meeting, the author **paraphrases** the **document** a **section at a time**



Responsibilities of Author

- Responsible for
 - **scheduling** the review
 - selecting the **review participants**
 - **determining** if the **entry criteria** for the review are met
 - providing **information** about the **product** during all stages
 - clarifying any **unclear issues**
 - correcting any **problems identified**
 - providing dates for **rework** and **resolution**



Responsibilities of Recorder



- Collects and records each defect uncovered during the review meeting
- Develops an issues list and identifies whose responsibility it is to resolve each issue
- Records meeting decisions on issues; prepares the minutes; and publishes the minutes, and continually tracks the action items

Responsibilities of Reviewer

- Spends time prior to the meeting **reviewing information**
- Makes **notes of defects** and becomes familiar with the **product** to be **reviewed**
- Identifies **strengths** of the **product**
- Verifies that the **rework is done**



Responsibilities of Observer

- A **new member** to the project team, who **learns** the **product** and observes the **review techniques**.



Review Guidelines

- Preparation
- Discussions
- Respect
- Agenda
- Review Records
- Resources
- Attendees

Review Frequency

- At the beginning/end of the requirements phase
- At the beginning/end of the design phase
- At the beginning/end of the code phase
- At the beginning/end of the test phase
- Approval of the test plan

Review Planning

- Distribute review package one week in advance
 - Document to be reviewed
 - Review agenda
 - Identification of the individual who will manage the agenda and schedule
 - Exit and entrance criteria for the review
 - Objective of the review
 - Names of attendees, their roles and responsibilities



Review Planning

- Distribute review package one week in advance
 - Review **location**
 - **Date** and **time** of review
 - **List** of classifications that will be used for **defects discovered** (defect type, defect origin, and defect severity)
 - Procedures for **handling issues** raised during the review



Review Meeting

- Facilitator begins the meeting with an **introduction** of **agenda**, **people**, and description of their **roles**
- Author of the document proceeds to explain the **materials**, while reviewers **raise issues** based on advance preparation



Review Meeting

- When valid **problems**, **issues**, or **defects** are discovered, they are **classified** according to their **origin** or severity and then recorded
- These are **accompanied** with the **names** of **individuals** who are **responsible for resolution** and the **time frame** during which the **item** will be **resolved**
- Related **recommendations** are also **recorded**



Decisions at the End of a Review Meeting

- All attendees must decide whether to
 - Accept the product without further modification
 - Reject the product due to severe errors
 - Accept the product provisionally
 - Hold a follow-up review session

Rework

- It is the responsibility of **project manager** to ensure that all **defects identified** in the review are **fixed** and **retested**

Follow-Up

- During the follow-up, that all **discrepancies identified** are **resolved** and the exit criteria for the review have been met
- Document **lessons learned** during the **final report** also

Task

- Facebook Login
- Overview of the Reviews process
- Roles



Recap

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